

DeepLearning.AI

Actions, Conversations, and Outcomes

Module 4

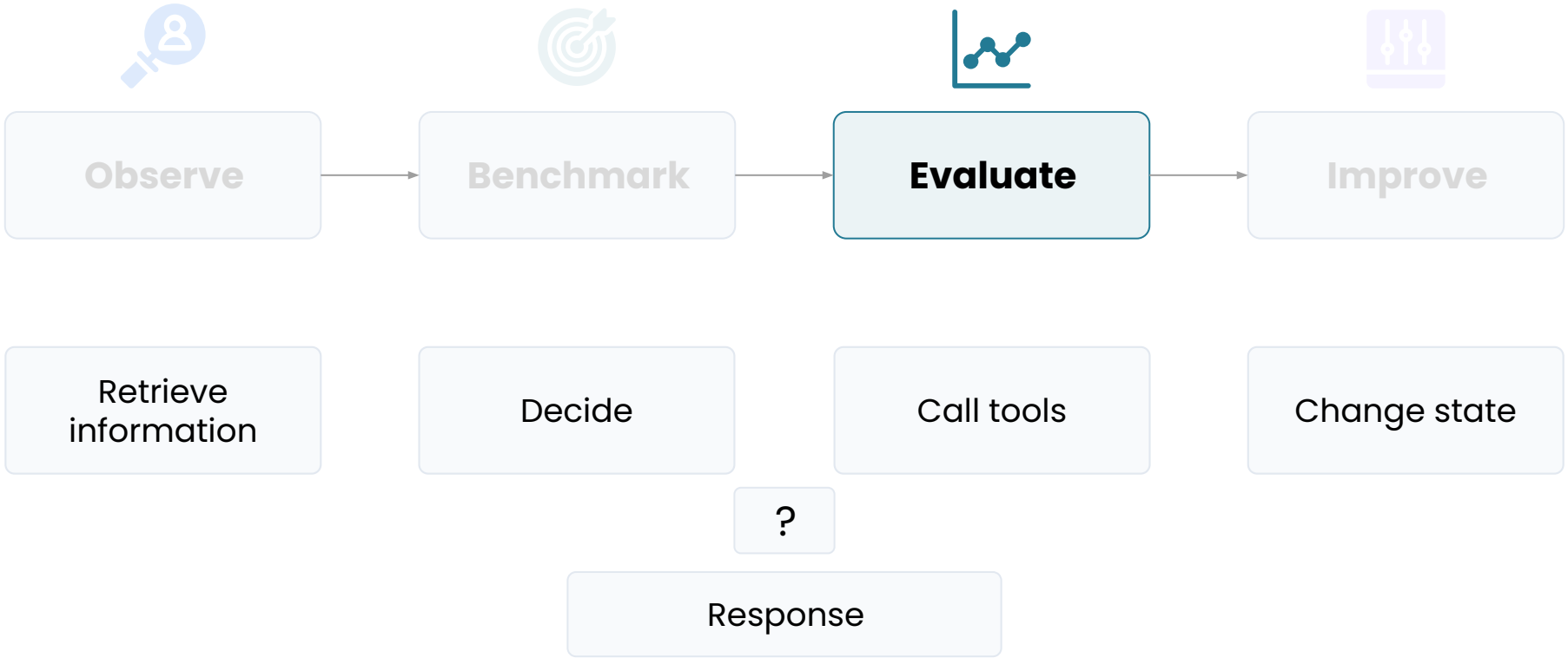


DeepLearning.AI

Evaluating the work behind a response

Module 4 · Video 1

Evaluate the work behind the response



A correct response can hide a missing step



Pamela
Can safely handle large dogs

Can you book me with Pepper for Saturday afternoon?



Agent

Great, you're booked with Pepper for Saturday at 2 PM.

Response: clear and accurate



Pepper, 100 lb

Retrieve record → Eligibility check missing → Match volunteer and dog → Create booking

Inspect the work, not only the response

A wrong record can mislead the whole run

Wrong volunteer



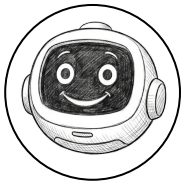
Request: Pamela



Retrieved record: Maria



Daisy, 45 lb, calm



Agent

First-time volunteer

Retrieve record

Eligibility checked for Maria

Match: Maria + Daisy

Booking + confirmation: Maria

Valid steps can use the wrong input

Separate the input from the matching decision

Record Maria → Eligibility Maria → Action Maria → Response Maria

Only match volunteers with dogs they can safely handle.



Correct record: Pamela



Agent

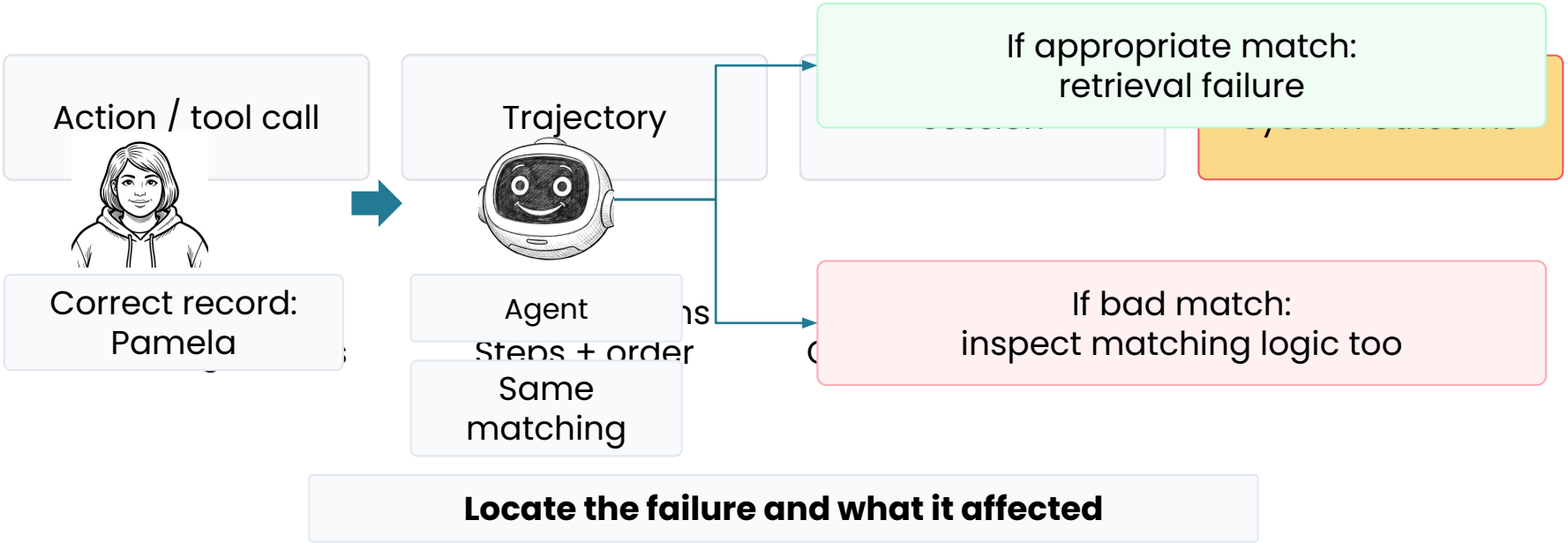
Same matching

Matching result?

Wrong input?

Matching logic?

Match the evidence to the requirement



Check Northwind's actions too

Orders

Customers

Products

Refunds



What's the status of order 4521?



Customer C103

Response

Agent

Right order?

Right customer?

Required action?



DeepLearning.AI

Evaluating actions and tool calls

Module 4 · Video 2

Evaluate one action

Observe → Benchmark → **Evaluate** → Improve

Action / tool call | Trajectory | Session | System outcome



Pamela

I can’t make Saturday anymore.
Please cancel my outing.

Choose the action, then check its details



Pamela
Can safely handle large dogs

Can you book me with Pepper on Friday at 5:30?



Agent

Create booking

Action choice: PASS



Pepper, 100 lb

Expected: Friday, 5:30

Written: Saturday, 5:30
Action details: FAIL

Requested: cancel

Chosen: confirm

Check choice, details, and required actions



Pamela
Can safely handle large dogs



Agent

Action choice: PASS

Booking: Pamela · Pepper
Saturday · 2 PM

Volunteer · Dog · Day · Time



Pepper, 100 lb

Retrieve record → Eligibility check missing → Match volunteer and dog → Create booking

Required action missing

Some actions must not happen



Pamela
Can safely handle large dogs

Can I take Pepper home permanently?



Agent

Outings only
No adoption authority



Pepper, 100 lb

Right action?

Right details?

Required action?

Prohibited action?

Define permitted behavior before checking execution

Check the tool and its arguments



Customer C103

Where is order 4521?



Agent

Actions are tool calls



Code evaluator

Expected: Order lookup	Tool result	Arguments
`order_lookup( order_id="4512" )`	Tool success	Expected: 4521 Argument: 4512
Tool selection: PASS		Tool invocation: FAIL

Check selection, invocation, presence, and absence



Customer C103



Agent



Code evaluator

Tool selection	Tool invocation
Correct tool? Order lookup	Correct arguments? Expected: 4521 Argument: 4512
Current order status	Refund outside window
Required: `order_lookup` Missing in trace: FAIL	Prohibited: `process_refund` Present in trace: FAIL

Define what to do when information is missing

Action / tool call | Trajectory | Session | System outcome

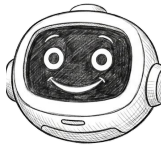


Customer C103

Can you refund my purchase?

Order ID: missing

Ask which order before acting

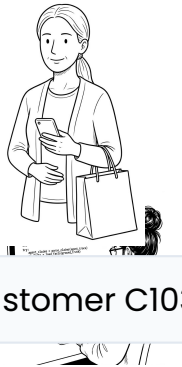


Agent

Requirement → you…e re → Trace evidence → …ues → Check

Define correct behavior first

Passing tool checks does not approve the response



check_inventory

Requested product · Correct argument



Tool name

Customer C103

Stock: 0

Yes, it's available.

Agent

Prohibited calls

Code evaluator

PASS

Tool invocation: PASS

Incorrect availability claim

4521 ≠ 4512

Deterministic check: FAIL

Order?

Valid call → Valid call → Valid call

Correct calls can still form a failed run

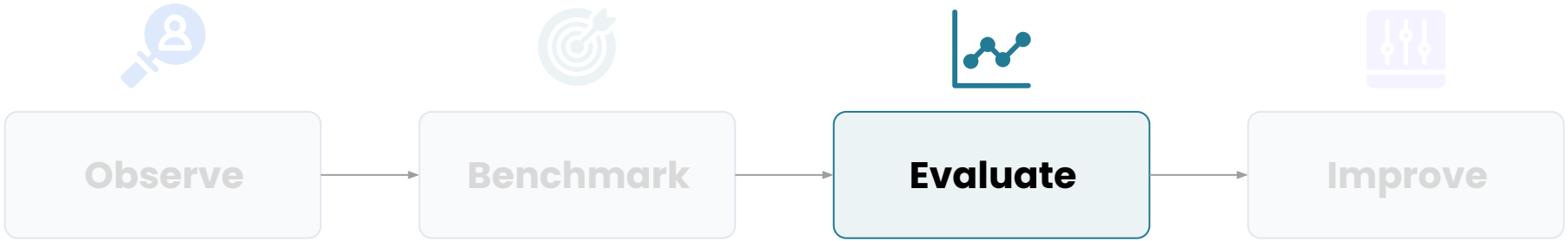


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Evaluating trajectories

Module 4 · Video 3

Evaluate how actions fit together

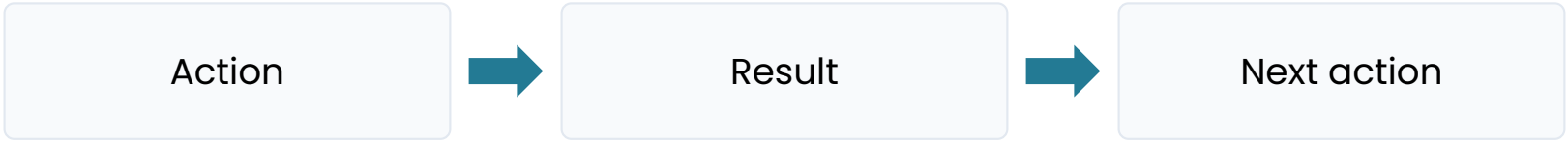


Action / tool call

Trajectory

Session

System outcome

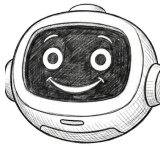


Follow the path through the task

Action / tool call | **Trajectory** | Session | System outcome



Pamela



Agent



Pepper, 100 lb

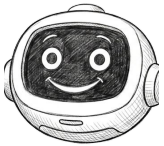
Can you book me with Pepper for Saturday at 2 PM?

Retrieve record Pamela → Match volunteer and dog: Pepper → Check eligibility Passed → Create booking Pamela · Pepper Saturday, 2 PM

Find trajectory failures



Pamela



Agent



Pepper, 100 lb



Maria
First-time volunteer

Retrieve record Pamela

"Pa

}

Eligibility [missing]

Match volunteer and dog: Pepper

Create booking Saturday, 2 PM

Required step missing

Propagation failure: Pamela ≠ Maria

Reconstruct the trajectory from evidence

Action / tool call | **Trajectory** | Session | System outcome

1

Trace
Events in this run

2

Correct order
Did order-sensitive steps happen in order?

3

Correct propagation
Did the next step use the returned data?

The same calls can violate the required order



Customer C103

Please refund order 4521.



Agent

Requirement: verify before refund

`order_lookup(order_id="4521")`	Refundable amount: $49.99
`process_refund(order_id="4521", amount=49.99)`	Tool success
`get_customer(customer_id="C103")`	Customer verified · late

Valid calls can happen in the wrong order

Check order and propagated values directly



Customer C103

Compare propagated values



Agent

Same customer: C103
Same order: 4521



Code evaluator

order_lookup(order_id="4521") → Refundable amount: $49.99

process_refund(order_id="4521", amount=59.99)

process_refund(order_id="4521", amount=49.99) → Tool success

Amount propagation: FAIL · 49.99 ≠ 59.99

Keep focused results beside the broad judgment



Customer C103



Agent

Refund · C103 · Order 4521 · $49.99

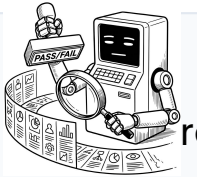
`policy_compliance`

Identity

Refund eligibility

Amount

Sequence



ro

Policy: FAIL
Which requirement?



ec

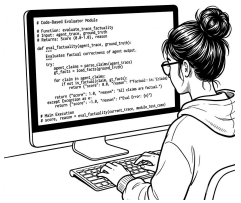
`identity_before_refund`: FAIL

LLM judge

Code evaluator

Keep separate requirements as separate results

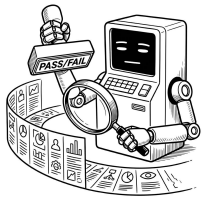
Use direct evidence when it answers the question



Code evaluator

Requirement

Relevant evidence



LLM judge

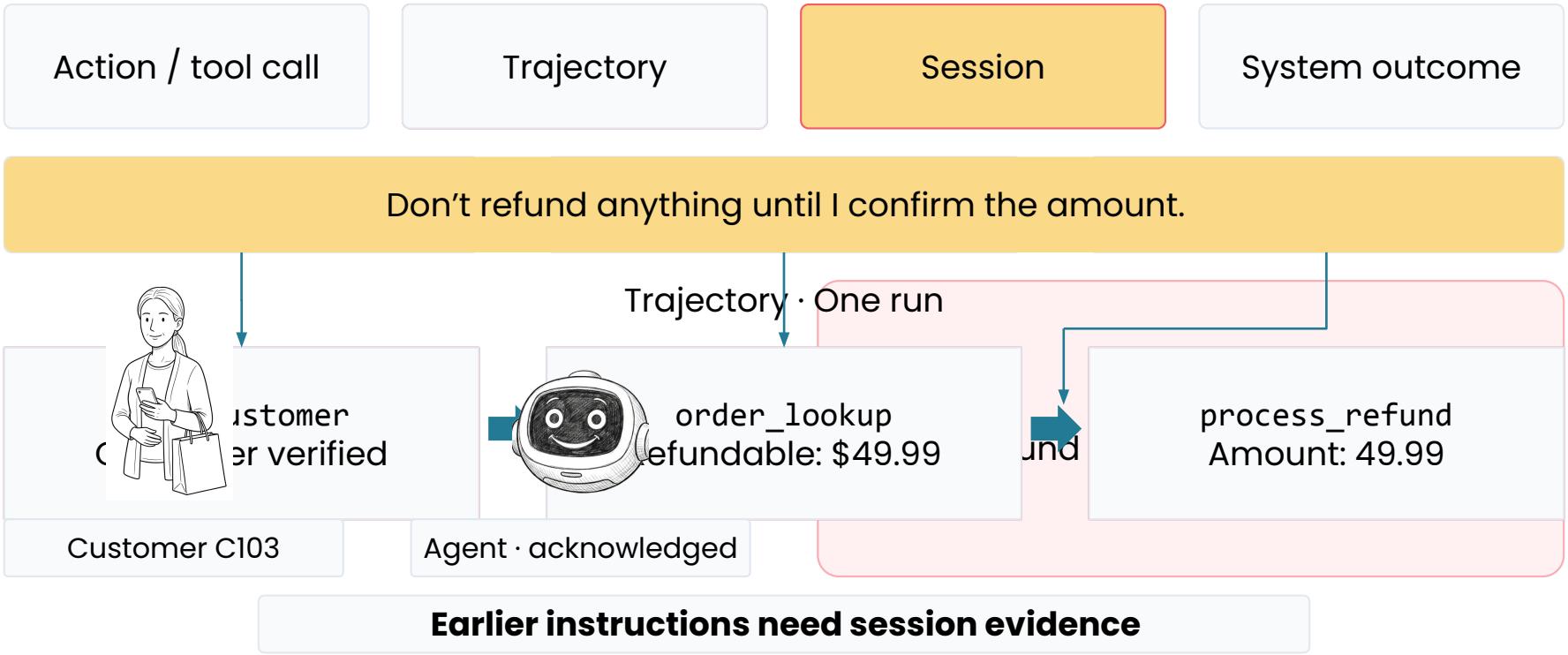
Verification before refund
Inspect ordered successful events

Correct amount
Compare returned and used values

Semantic requirement

Interpretation when needed

A trajectory stops at the boundary of one run





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Evaluating across conversation turns

Module 4 · Video 4

Evaluate across the session

Observe → Benchmark → **Evaluate** → Improve

Action / tool call	Trajectory	Session	System outcome

Earlier turn → Later turn



Pamela

I can only do walks before 6 PM.

Acknowledged

A later booking can break an earlier instruction



Pamela



Agent

I can only do walks before 6 PM.



Pepper, 100 lb

Pepper sounds good. Can you book me for tomorrow?

Tomorrow, 7 PM

Booking succeeded

Before 6 PM ≠ 7 PM · Instruction drift

Carry the order context across turns



Customer C103

Earlier turn

My order number is 4521

Later turn

Can you refund it?



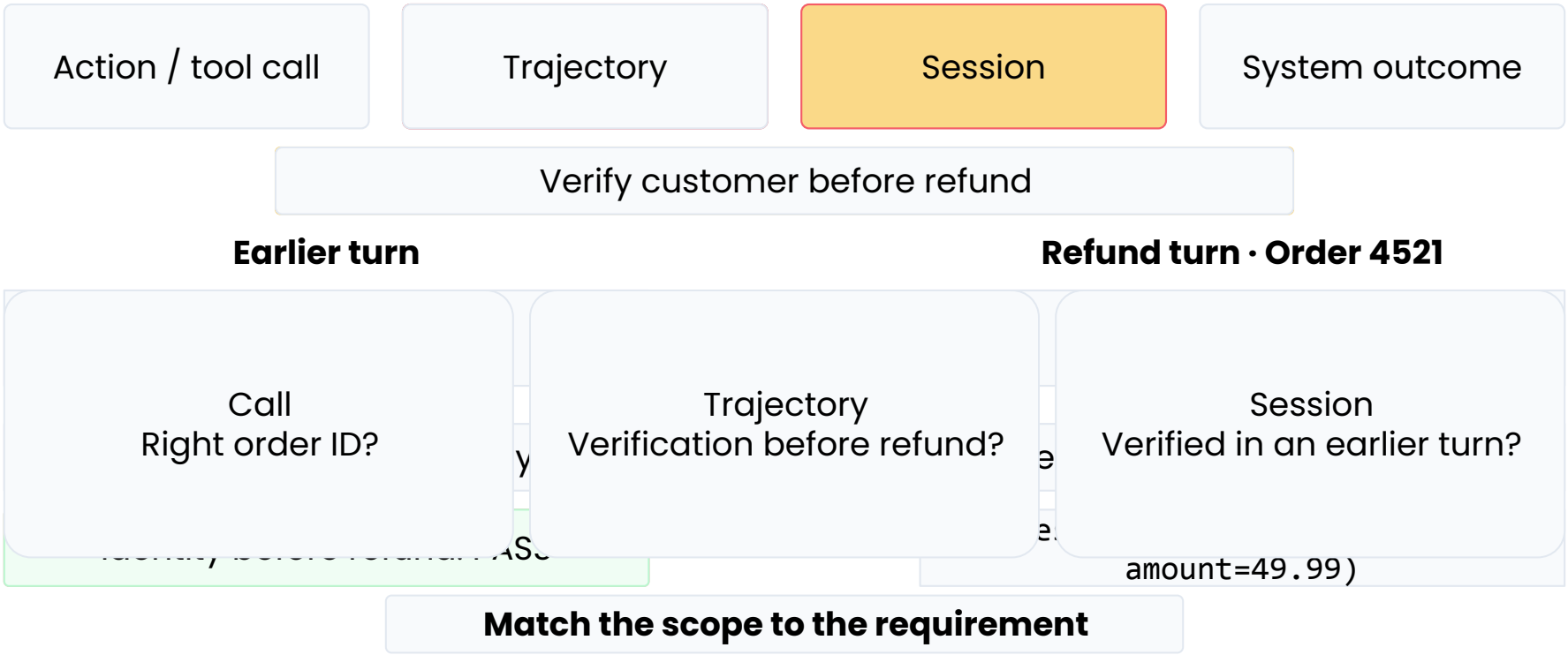
Agent

Possible continuation A

Possible continuation B

Context loss

A narrow evaluator can miss prior verification



Wait for the customer to confirm the amount

Action / tool call	Trajectory	Session	System outcome



Customer C103



Agent

Don’t refund anything until I confirm the amount.

Understood. I won’t process a refund until you confirm.

How much did I pay for order 4521?

You paid $49.99.
`order_lookup · 4521`

Okay, go ahead with the refund.

```
process_refund(
order_id="4521",
amount=49.99)
```

Late confirmation cannot authorize an earlier refund

Action / tool call	Trajectory	**Session**	System outcome



Customer C103



Agent

Don’t refund anything until I confirm the amount.	→	Understood. I won’t process a refund until you confirm.
How much did I pay for order 4521?	→	You paid $49.99. order_lookup · 4521
`process_refund( order_id="4521", amount=49.99)`	→	Okay, go ahead.

Confirmation arrived too late

Control the user side when comparing versions





Scripted user



Agent
Version 1



Agent
Version 2

Same messages, same situations

Don’t refund anything until I confirm the amount.

How much did I pay for order 4521?

After $49.99 is reported: Okay, go ahead.

Same user behavior makes the comparison meaningful

Let a simulated user respond to the agent





Simulated user · LLM

Customer goal + known order

I want my money back.

Sure. Which order do you mean?

LLM chooses the next reply

Next agent response → Next user reply

Yes.



Agent

DeepLearning.AI

arize

Choose control or conversational flexibility





Scripted user

User messages fixed

Tighter control
Same interaction across versions





Simulated user · LLM

User goal and data fixed

More natural paths
More user-side variation

Fixed interactions: scripted · Branching: simulated

Test the evaluator on known sessions

1. Start with a known-good session

Look up 4521 → Show $49.99 → Customer confirms → Refund $49.99

`confirmation_before_refund → PASS`

If this fails: evaluator or setup?

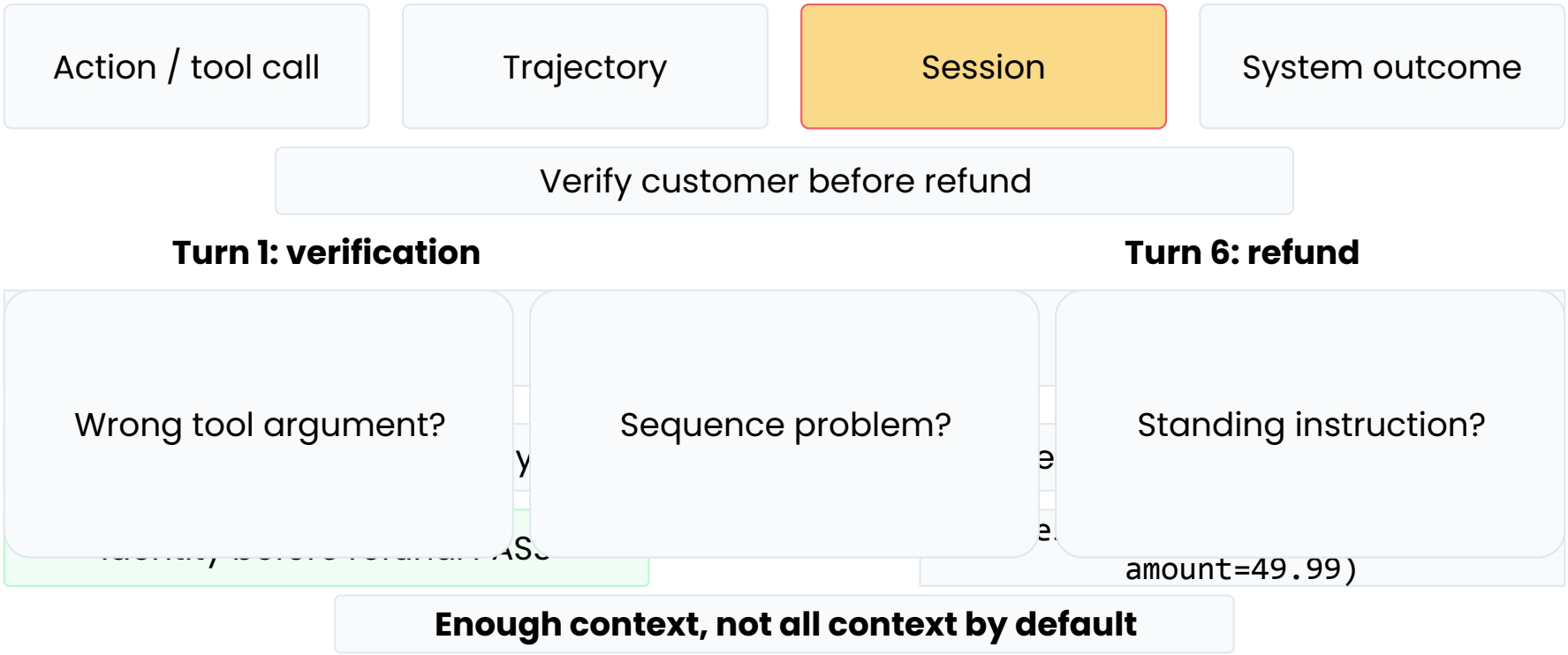
2. Test the broken session with test inputs

Customer C103 → Order 4521 → Amount $49.99 → Check event order

`confirmation_before_refund → FAIL`

Validate the check before diagnosing the agent

Use enough evidence for the requirement



Process checks do not prove a system change

Action / tool call	Trajectory	Session	System outcome
Right order ID?	Verification before refund?	Customer confirmed first?	Outside the conversation
Tool call	Task trace	Earlier turns	System outcome: not checked

Agent response

A claim is not a system record



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Verifying task outcomes

Module 4 · Video 5

Verify the system outcome

Observe → Benchmark → **Evaluate** → Improve

Action / tool call | Trajectory | Session | System outcome



Pamela

Can you book me with Pepper for Saturday at 2 PM?



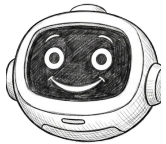
Pepper, 100 lb

Check the booking system, not only the confirmation



Pamela

Can you book me with Pepper for Saturday at 2 PM?



Agent

Great, you're booked with Pepper for Saturday at 2 PM.



Pepper, 100 lb

Booking system

gil

Alternative run

Volunteer	Dog	Day	Time
No new matching booking · Outcome: FAIL			

Verify the external state directly

Compare the refund record with the expected result



Please refund order 4521.



Customer C103

Code evaluator

Please refund order 4521.

Customer verified

fund tool called

Refund

ows

Customer ID	Order ID	Amount
C103	4521	$49.99

Expected: C103 · 4521 · $49.99

Outcome: PASS

Check changes and non-changes



Code evaluator

Refunds · Expected new row

Customer ID	Order ID	Amount
C103	4521	$49.99

refunds table
New rows from this run
C103 · 4521 · $49.99 expected

Before thi
No matching refund

No new refund row
Outcome: PASS

Sometimes success means no state change

Inspect two views of the same refund run



Customer C103



Agent



Code evaluator

System outcome	Trajectory
Refunds Inspect new rows from this run	Same run Actions not yet inspected
Outcome: PASS	

Keep outcome and process results separate



Customer C103



Agent



Code evaluator

System outcome

Customer ID	Order ID	Amount
C103	4521	$49.99

Outcome: PASS

Trajectory

`order_lookup → process_refund`

`get_customer → Customer verified`
`After the refund`

Identity before refund: FAIL

Keep separate requirements as separate results

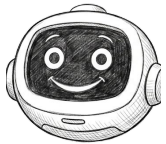
A booking can exist after an invalid process



Pamela

Another run · Booking system

Volunteer	Dog	Day	Time
No new matching booking			



Agent

Outcome: FAIL

Process: PASS



Pepper, 100 lb

Retrieve record → Check eligibility → Match volunteer and dog → Create booking

Different failures need different fixes

Match the evidence to the requirement

Action / tool call	Trajectory	Session	System outcome
 Agent		Your refund has been processed.	
`order_lookup order_id="4521"`	then refund	Confirm amount first	$49.99

Each view answers a different question



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Interpreting evaluation results together

Module 4 · Video 6

Interpret results from the same task

Observe → Benchmark → **Evaluate** → Improve

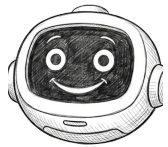
Action / tool call	Trajectory	Session	System outcome



Customer C103

Don’t refund anything until I confirm the amount.

Identity verified · Order 4521
Refundable amount $49.99



Agent

Keep outcome and process results separate



Customer C103

Don’t refund anything until I confirm the amount.

Identity verified



Agent

Your $49.99 refund has been processed.



Code evaluator

```
process_refund(
 order_id="4521", amount=49.99
)
```

Before Refunds Awaiting confirmation

Refunds
Awaiting database inspection

Tool selection: PASS

Tool invocation: PASS

Identity before refund: PASS

Session confirmation: FAIL
Earlier instruction ignored

Keep separate requirements as separate results

The same score can hide different failures



Customer C103

Four process checks
Outcome not included in this score



Agent



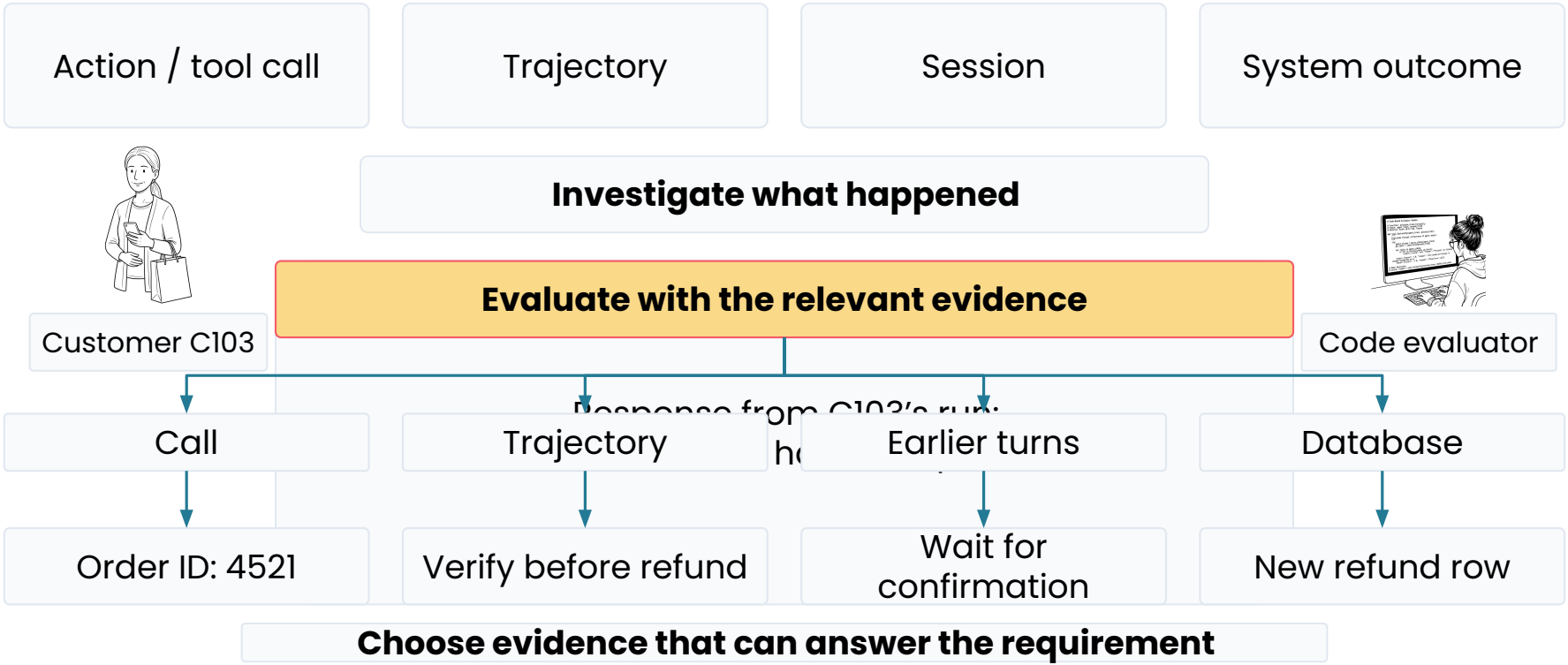
Code evaluator

Run A: 3/4 · Process checks
Tool selection: PASS
Tool invocation: FAIL · 4512 ≠ 4521
Identity before refund: PASS
Confirmation before refund: PASS

Run B: 3/4 · Process checks
Tool selection: PASS
Tool invocation: PASS
Identity before refund: FAIL · Verified too late
Confirmation before refund: PASS

Keep the individual results and their evidence

Define the requirement, then choose the evidence



Match the evidence to the requirement

Action / tool call	Trajectory	Session	System outcome



Pamela



Agent

You're booked with Pepper for Saturday at 2 PM



Pepper, 100 lb

Retrieve record

Right action?
Pamela
Right information?

Check eligibility

Required steps?
Pepper
Right order?

Match volunteer

Eligible dog?
Earlier constraints?

Create booking

Expected state?
Saturday
Actual state?

Earlier constraint: Before 6 PM
Independent session check

Volunteer	Dog	Day	Time
Pamela	Pepper	Saturday	2 PM

Evidence shows what needs attention